

# Debapratim Deka

✉ debapratimdeka2001@gmail.com 📞 +91 8638454246 🔗 LinkedIn 🌐 GitHub 🌐 Portfolio

**Data Scientist & AI Engineer** with **4+ years** of experience architecting and productionizing **ML, GenAI, LLM, and Agentic AI** solutions across enterprise use cases, specializing in **RAG, multi-agent systems, NL-to-SQL, and AI automation**. Experienced in **AI solution architecture, technical leadership, and MLOps**, delivering scalable production systems using **Python, Databricks, Azure, Kubernetes, CI/CD, LangChain, LangGraph, and Azure OpenAI**.

## Work Experience

### Sigmoid Analytics - Bangalore

#### Associate Lead Data Scientist

Jan 2026 - Present

##### • AI-Powered Data Sufficiency & Governance Platform – US Consumer Health Company:

- **Architected and led a 3-member team** to productionize an enterprise **AI-powered Data Sufficiency platform**, owning solution architecture, technical development, stakeholder alignment, and end-to-end delivery across UC Hydration, AI Summary, Entity Resolution, and Scoring services.
- **Designed and developed an agentic AI framework** to extract and enrich enterprise data context including **metadata, lineage, KPIs, business definitions, relationships, dimensions, and granularity**; built a **data-readiness scoring engine** mapping business use-case requirements to available data and generating suitability scores, summaries, and heatmaps for identifying data gaps and readiness.
- **Productionized the AI platform** using cloud-native engineering practices with containerized services on **Kubernetes/AKS, GitHub-based CI/CD, automated deployment, scalability, monitoring, and observability**, enabling reliable enterprise AI workloads.

#### Senior Data Scientist

Jan 2025 - Dec 2025

##### • Scientific Claims Strategy & Competitive Landscaping – US Consumer Health Company:

- Curated an **Ingredient+Literature Knowledge Base** by mining scientific databases and competitor systems, reducing manual sourcing effort by **40%**; developed hybrid **RAG & CAG** with Claims Bracketing and Ingredient Equivalency methodologies, driving a **20% revenue** surge.
- **GenAI Executive Summary & Multi-Agent Analytics** – US Consumer Health Company:
  - Architected a **multi-agent GenAI analytics platform** across **50+ enterprise tables** spanning Sales, Finance, and Commercial domains, enabling automated executive summaries and context-aware analytics with **NL-to-SQL, query validation, KPI/entity extraction, SQL self-correction, dynamic visualization, and domain-aware memory**; **reduced analysis turnaround by >99%** and achieved **87% chatbot accuracy**.
- **Figma-to-React Code Generation**:
  - Architected an **agentic AI platform** converting Figma screenshots and multi-page PDFs into production-ready **React TypeScript applications**, leveraging **LLM routing, UI rule books, multi-page context, code generation/modification, and critic-build validation loops**; achieved **93% average visual similarity** while enabling natural-language codebase modification.

#### Data Scientist

Jan 2024 - Dec 2024

##### • Claim Generation Assistant – US Consumer Health Company:

- Engineered a **GenAI** pipeline integrating LLM sentiment analysis and social listening, processing **10M+** reviews and accelerating claim generation by **30%**; built a claim recommendation assistant using fine-tuned **T5, BART, and GPT**, achieving **85% accuracy** and compressing brand insights by **75%**.
- **Sales Driver Attribution Analysis** – US Alcoholic Beverages Company:
  - Implemented a **constraint-based regression** model identifying sales depletion drivers across operations, macro trends, and weather data, achieving **89% accuracy**; delivered predictive dashboards quantifying pricing, promotion, and media impact across **50 US states** for sales and inventory optimization.
- **Forecast Failure Analyser** – US Food Company:
  - Developed a **GenAI-driven** forecasting diagnostic solution using **LangChain, LangGraph, ReAct, and FastAPI** across **100K+ SKUs**, improving diagnostic speed by **65%**; enabled 30+ analytics tasks, accelerating reporting by **60%** and improving LLM performance by **90%**.

#### Associate Data Scientist

July 2022 - Dec 2023

##### • Supply Chain Forecasting – US Food Company:

- Constructed scalable forecasting pipelines for **100K+ SKUs** using Fourier transforms, Y-lags, and segmentation, boosting performance by **10.6%** and achieving **82% system accuracy**.
- Optimized models using **Optuna, Bayesian optimization, and volatility-based diagnostics**; developed trend indicators in Databricks pipelines, boosting precision by **40%** across high-variance segments with Snowflake integration.
- Developed **demand-type-specific models** with classification routing and **VAR-based pipelines**, leveraging inter-SKU dependencies and **Granger causality** for lag optimization across diverse demand patterns.
- Devised a **hybrid QLinEx loss function** with asymmetric error penalization for constrained forecasting, improving regression fidelity by **15%** across volatile product lines.

## Skills

- **AI / GenAI / LLM**: Generative AI, Agentic AI, Multi-Agent Systems, LLMs, RAG, CAG, Prompt Engineering, NL-to-SQL, Text-to-SQL, LLM Orchestration, AI Code Generation, LLM Evaluation, NLP, T5, BART, GPT, Azure OpenAI, LangChain, LangGraph, ReAct, Vector Databases, Graph Databases, Knowledge Graphs
- **Machine Learning**: Supervised & Unsupervised Learning, Deep Learning, Time Series Forecasting, Regression, Classification, Clustering, Feature Engineering, Hyperparameter Optimization, Bayesian Optimization, Statistical Modeling, Causal Analysis, Optimization Techniques, Model Evaluation
- **AI Engineering / MLOps**: FastAPI, MLflow, Kubernetes, AKS, Docker, GitHub, GitHub Actions, CI/CD, Helm, Argo CD, Cloud Deployment, Monitoring, Observability
- **Data Engineering / Cloud / Databases**: Python, PySpark, SQL, MySQL, PostgreSQL, Snowflake, Databricks, Azure, AWS, Data Pipelines, Metadata Management, Data Lineage, Data Governance, Vector DB, Graph DB
- **Libraries / Frameworks**: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, Statsmodels, NLTK, TextBlob, spaCy, TensorFlow, Keras

## Education

### NATIONAL INSTITUTE OF TECHNOLOGY SILCHAR

B.Tech, Electrical Engineering

July 2018 - July 2022

CGPA - 9.49

## Awards

- Received **three consecutive SPOT Awards** (Q1 2023 – Q3 2024) for excellence in scalable forecasting, GenAI-powered assistants, and delivering innovative, high-impact data science solutions.